

Theorem 2.8 Properties of Exponential Functions

Let x, y be real numbers. Let $a, b > 0$

Rule	Formulation
Equality Rule	$b \neq 1$ implies $b^x = b^y \Leftrightarrow x = y$
Inequality Rules	Case $b > 1$ $x < y \Rightarrow b^x < b^y$ Case $b < 1$ $x < y \Rightarrow b^x > b^y$
Product Rule	$b^x b^y = b^{x+y}$
Quotient Rule	$\frac{b^x}{b^y} = b^{x-y}$
Exponent Rule	$(b^x)^y = b^{xy}$
Power Rules	$(ab)^x = a^x b^x$ $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
Special Values	$b^1 = b, \quad b^0 = 1$